

# Shortcut management strategy evaluation: short-run or short-changed

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Manuscript from August 13, 2026

*Note: editorial notes in italics and red (remove before submission)*  
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## 1 Abstract

to do (later)

*Keywords:* management strategy evaluation, shortcut, simulation testing, fisheries management

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### 3 Introduction

#### *MSE intro (concept, OM & MP, history, ...)*

Management strategy evaluation (MSE) describes the process of testing fisheries management strategies (MS) with computer simulations to provide a quantitative means to evaluate the performance of candidate MSs against management objectives while taking cognisance of key uncertainties in the underlying resource(s) dynamics (Punt et al., 2016). The crucial step in MSE is that simulations include a feedback loop (also known as closed loop), which distinguishes it from other stochastic projections without feedback (i.e. open loop). The concept of simulation testing has been around for decades and was pioneered at the International Whaling Commission (IWC) in the late 1980 (Butterworth, 2007) but had been used previously, e.g. for Pacific halibut (Southward, 1968). The term MSE itself was introduced in the 1990s (Smith, 1994) when it was rolled out more widely, e.g. in South Africa (Butterworth & Bergh, 1993).

MSE is ideally a process that involves iterative interactions between analysts and stakeholders, but can also be limited to a closed-loop simulation exercise. To distinguish between different MSE nuances, Walter et al. (2023) defined the terms “desk MSE”, “intermediate MSE”, and “full stakeholder MSE”, where the first refers to MSE simulation exercises and the latter includes the full process, including iterative stakeholder-analyst interactions.

Punt et al. (2016) define seven steps required for an MSE: (1) identification of management objectives, (2) identification of uncertainties, (3) development and (4) conditioning of operating models (OM) to data, (5) identification of candidate management strategies (MS), (6) simulation of each MS for each OM, and (7) summarise, interpret results, and chose a best performing MS. The OM represents the biological population, the fishery, and the data collection. In an MSE, data generated in the OM is then passed to the MS. The MS is a pre-specified algorithm, which can include an estimation model, such as a stock assessment model, and a decision method to derive a management action, such as a catch limit (total allowable catch, TAC). MSs which include an estimation model are called “model-based” (Figure 1a), while those which do not are called “empirical” or “model-free” (Figure 1d). In an empirical MS, the decision method is a pre-specified algorithm (harvest control rule, HCR) that depends on the observed data but does not model the resource dynamics. In contrast, in a model-based MS, the data are used in an estimation model (stock assessment) to model the resource, and the output of the estimation model is passed to the decision method. What sets an MSE apart from other forward simulations is that only data and no other metrics are passed from the OM to the MS, and that the output from the estimation model is returned to the OM in a feedback loop (Figure 1a). A fully specified MS that has been simulation tested and approved by MSE is called a management procedure, the component of which represents the collection and analysis of data, the assessment method (or empirical rule), and the HCR to set catch advice.

An MSE can be based on a single (baseline) OM, which represents the most plausible representation of reality. However, it is useful to include several alternative OMs that represent different scenarios, such as different productivity regimes, or structural uncertainties that represent uncertainty about resource dynamics (Punt & Donovan, 2007). These are typically grouped into a reference set of more plausible scenarios and a robustness set of more extreme scenarios (Punt et al., 2016). In general, the complexity of an MSE should be in the OM or a set of OMs to thoroughly evaluate the performance and robustness of an MS. MSE can be used to test an MS’s performance (i.e. how well it meets robustness of desired objectives such as maximising catch or reducing the risk of stock depletion), to compare different candidate MSs, or to tune an MS. In this context, tuning refers to the process of adjusting the control parameters of an MS to find those that best meet objectives, which can be computationally demanding because many control parameters and combinations may need to be explored. If an OM reference set is used, performance is measured across the set, and the entire set is used for the tuning.

It is best practice to test the MS and all of its components in an MSE as they would be applied in reality (Punt et al., 2016). The estimation model may have high computational complexity, such as an integrated stock assessment model combining data from multiple fisheries or fleets. Running the estimation model thousands of times in an MSE requires substantial computing resources and runtime. In addition, achieving acceptable convergence with all assessments can be difficult to automate. This computational challenge is likely the primary reason for the emergence of “shortcut” approaches that simplify or remove the estimation model. However, to the authors’ knowledge, no study has yet addressed the definition of shortcut MSE or its potential dangers.

It is important to point out that an MSE with an empirical MS is not a shortcut MSE, because the MS, by design, has no estimation model, and the MS is tested in the MSE as it is applied in practice, which should include any processing of observations, e.g. to develop indices of abundance.

The objectives of this paper are threefold:

1. First, we define the different types of shortcut MSE.
2. Second, we take several case studies for which a full MSE has previously been conducted and published and apply a shortcut MSE, thereby allowing us to compare the two approaches.
3. Thirdly, we objectively discuss the advantages and disadvantages of the shortcut approach, based on the case study results and theoretical considerations, and make a recommendation on whether and how shortcut MSE should be used.

## 4 Shortcut MSE definition

*Note: content/text copied and adapted from ICES MSE course slides*

A shortcut MSE is a type of MSE where the MS is simplified, either by simplifying the estimation model (stock assessment model) or by replacing it altogether with an estimation emulator. For clarity in this study, we refer to non-shortcut MSE as a “full MSE”. We define three types of shortcuts, loosely based on discussions at an ICES workshop (ICES, 2020a):

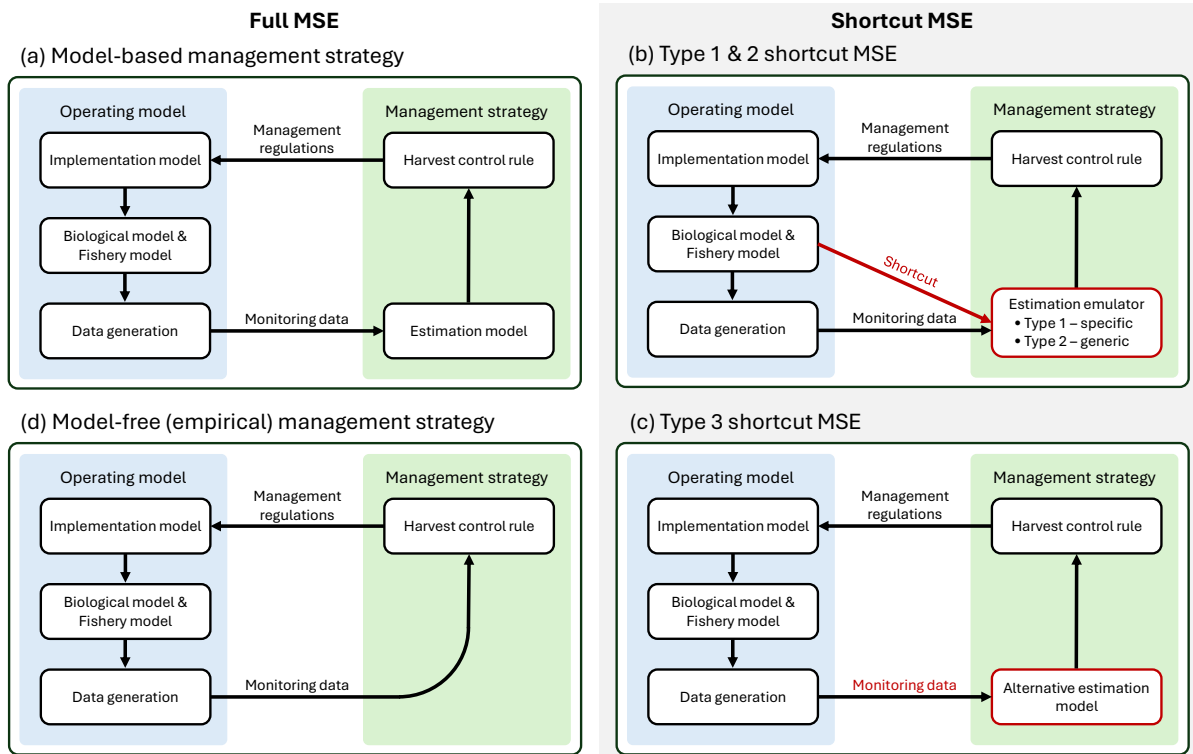


Figure 1: Illustration of management strategy evaluation (MSE, a) and the three shortcut MSE types (b–c). The design for a model-free (empirical) management strategy is shown for comparison (d), but it is not a shortcut.

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- **Type 1 shortcut MSE – specific emulator**

Type 1 is defined by the estimation model (stock assessment model) being replaced entirely by an estimation emulator, which is meant to emulate the behaviour of the estimation model. This is typically done by taking the numbers-at-age or biomass directly from the OM and adding noise (estimation error) before passing it to the decision model. This can be done by using the assessment error, e.g. from the variance-covariance matrix. Alternatively, an analytical retrospective analysis can be conducted with the estimation model (i.e. using real historical data and removing data from the end and re-running the estimation model outside the MSE loop), and the retrospective pattern can be used to define an estimation error, including auto-correlation if necessary. With the increased availability and development of machine learning and artificial intelligence, more complicated estimation emulators may evolve. Depending on the design of the MS, the data generation module of the OM may be removed if it is not required; however, some MS may use, e.g. a

short-term forecast and some monitoring may still be required.

- **Type 2 shortcut MSE – generic emulator**

The design of a type 2 shortcut is the same as that of a type 1 shortcut MSE, but the estimation emulator is generic, not conditioned on a specific estimation model, and not intended to emulate the behaviour of one particular estimation model. This is typically done by adding random noise to OM metrics. This type of shortcut follows a looser interpretation of MSE, focusing on evaluating HCRs (e.g. different functional forms) but without pre-specifying the monitoring data or assessment method. If applied to a specific stock, the estimation emulator noise for this shortcut could come from a historical retrospective pattern, i.e. by comparing results from successive historical stock assessments. This differs from an analytical retro because the estimation model or its configuration may have changed over time, or historical data may have been revised. This type of shortcut cannot be tested against a full MSE because the estimation model is not defined.

- **Type 3 shortcut MSE – simplified estimation model**

Type 3 is characterised by replacing the original estimation model with a simpler estimation model. This may be needed because the original estimation model is too computationally demanding and cannot be included in the simulation. A simpler estimation model can be a different stock assessment model of the same complexity (e.g. another stochastic age-structured model) that runs faster, or a structurally simpler estimation model (e.g. replacing an age-structured model with a biomass-dynamic model). Care should be taken because different estimation models may behave differently, particularly under changing conditions during an MSE projection. For this type of shortcut, data generation in the OM is still needed, but may be adapted to the requirements of the estimation emulator. While candidate MSs with simpler estimation models are generally encouraged, the reason this is considered a shortcut is that the MS to be implemented in practice would use a different (more complicated) estimation model than that used during the MSE.

In addition to the three shortcut MSE types described above, there are other ways to simplify MSE. Most modern stock assessment models, whether they are state-space models such as SAM (Nielsen & Berg, 2014), integrated models such as Stock Synthesis (SS3; Method & Wetzel, 2013) or Bayesian models such as Just Another Bayesian Biomass Assessment (Winker et al., 2018, JABBA;), include options to reduce their runtime. Options to include this are relaxing convergence criteria (i.e. the model converges earlier), providing initial parameters from a previous year of the MSE simulation, turning off uncertainty estimation, or simplifying the model or data structure by removing or combining fleets with little impact on the outcome. For example, ICES (2019) used SAM in the MS but reduced the runtime to less than 10% by relaxing the convergence tolerance, removing Newton steps, and providing more informative initial parameter estimates. The Surplus Production Model in Continuous Time (SPiCT Pedersen & Berg,

2017) typically runs with several time steps per year, but these can be removed for speed.

A crucial issue with shortcut MSE approaches is how to deal with alternative OM. It may be feasible to condition an estimation emulator on a baseline OM, but estimation models may behave differently if metrics such as natural mortality or total catch levels differ. In practice, the only input to MSs are monitoring data. One of the primary advantages of MSE is that it can be used to develop MSs, which are shown to be robust to key uncertainties. These uncertainties are not only simulated within each OM, but also through different key assumptions in alternative OM. In full MSEs, candidate MSs respond to monitoring data only, as would occur in practice. However, in a short-cut MSE (types 1 and 2), the estimation emulator implicitly knows about any changes in the OM because it directly receives the information about the biology and fishery from the OM and the MS can directly respond to that information. This means that in a full MSE, misspecifications in the estimator can be tested, for example, when “true” natural mortality or total catch levels in the OM differ from those assumed by the estimation model. However, when an estimation emulator is used, the MS will implicitly “know” about any changes in the “true” dynamics, and it is therefore impossible to properly test any misspecifications.

## 5 Use of shortcut MSE in ICES

### *some examples and recent developments*

The International Council for the Exploration of the Sea (ICES) provides scientific advice on fishing opportunities for many fish stocks in the Northeast Atlantic (ICES, 2023a) and regularly conducts full and shortcut MSEs. Opinions on the use and appropriateness of shortcut approaches differ widely in the ICES community, and this is one of the reasons for the present study.

The issue of shortcut MSEs has been addressed several times over the past 20 years in ICES. Already in 2008, ICES (2008) compared the results of a full MSE for a North Sea cod recovery plan to a type 1 shortcut MSE and found that short-term performance was similar, but the long-term performance differed, and the shortcut MSE led to a different ranking of MS and would therefore lead to a recommendation of a different MS compared to conducting a full MSE. ICES (2008) concluded that full MSEs should be performed instead of shortcuts. The topic was later taken up again during ICES workshops on MSE guidelines (ICES, 2020a). During (ICES, 2020a), it became apparent that there were different interpretations of what a shortcut MSE is, something we define as type 1 and type 2 shortcuts. In the past, most MSEs in ICES were based on shortcuts, but this has changed in recent years (ICES, 2018, 2020a). There are now increasingly more full MSEs, e.g. ICES (2019, 2020b, 2024a, 2025a,c), but there are still some shortcut MSEs being conducted, e.g. ICES (2023c,b, 2025d)

Over the past years, management reference points have been calculated with a simulation tool called “EqSim” (Simmonds et al., 2022) in ICES (ICES, 2021). EqSim uses

long-term stochastic forward projections and is usually run without feedback (i.e. not an MSE) but can include feedback in the form of a very simple type 2 shortcut MSE. The approach of using EqSim for the provisioning of reference points has attracted criticism over the years because of its simplistic design, its failure to account for feedback between management and the biological population, and its disregard for the time lags between available data and management actions (Winker et al., 2025). There have been a series of workshops to advance reference point estimation (ICES, 2022a,a, 2024b) culminating in a new simulation framework that follows a simple type 2 shortcut MSE and is based on the Fisheries Library in R (FLR; Kell et al., 2007) framework and called Reference Point Estimation Tool (RPETool) by Winker et al. (2025). This tool further simplifies the type 2 shortcut by using the point estimates from the stock assessment model output as the deterministic starting point for the projections. Although the tool has existed for a few years, it has not been adopted by ICES, and reference points are still widely estimated using EqSim, despite evidence from ICES workshops that reference points may not be precautionary. A recent study from Winker et al. (2025) strengthens this perception by showing that EqSim reference points are not precautionary when compared to the new FLR shortcut MSE. The same FLR shortcut MSE framework had already been applied a few years ago to a Northern shrimp (*Pandalus borealis*) stock (ICES, 2023c). The use of the framework was possible in this case because it was a response to an EU request to evaluate a long-term management plan. Furthermore, this stock is assessed with an ensemble of three SS3 models with different natural mortalities, and the shortcut MSE estimation emulator was conditioned on the ensemble estimation model.

## 6 Case studies

This study presents several case studies to compare the outcomes of full and shortcut MSEs. The case studies were based on stocks for which ICES previously conducted full MSEs ICES (2019, 2025a) and we use their results. Subsequently, we repeated these MSEs but with a shortcut approach.

### 6.1 Western English Channel plaice

#### 6.1.1 Methodology

In 2024, ICES conducted a full MSE for European plaice (*Pleuronectes platessa*) in the Western English Channel as part of a benchmark workshop (ICES, 2025a) and full details of the MSE are described in Fischer (accepted). While this MSE focused on another management strategy (a model-free empirical HCR), the ICES MSY rule was also run and later tuned by Fischer (accepted). This consists of conducting an age-structured stock assessment (here, the state-space assessment model, SAM; Nielsen & Berg, 2014). The ICES MSY rule (ICES, 2023a) is a decision model which includes a harvest control rule of the form



$$F_{y+1} = F_{\text{target}} \times \min \left( 1, \text{SSB}_{y+1}^{\text{est}} / B_{\text{trigger}} \right), \quad (1)$$

where  $F_{\text{target}}$  and  $B_{\text{trigger}}$  are control parameters, and  $\text{SSB}_{y+1}^{\text{est}}$  the spawning stock biomass estimated by the stock assessment model at the beginning of year  $y+1$ , the year for which advice is given.  $F$  is subsequently converted into a catch value with a stock assessment-internal stochastic short-term forecast.

The plaice MSE included 15 OMs, split into a reference set (7 OMs) and a robustness set (8 OMs), which included alternative assumptions about catch levels (discard survival scenarios, OM1–2; alternative stock definition based on catches, OM6; implementation error, OM4–5), natural mortality (OM7–9), recruitment (auto-correlation, recruitment failure, higher/lower recruitment and steepness, OM10–14), and observation error (OM15). Each OM had 1,000 simulation replicates. The ICES MSY rule was tuned adjusting  $F_{\text{target}}$  and  $B_{\text{trigger}}$  and using standard ICES management objectives, i.e., finding the parameterisation that maximises the long-term catch (last 10 years of a 20-year projection while remaining precautionary. ICES considers management plans precautionary if the probability of the SSB falling below the biomass limit reference point  $B_{\text{lim}}$  (called  $B_{\text{lim}}$  risk here) does not exceed 5% in any year of the period considered. For plaice, the tuning was performed with OM reference set, i.e. by combining 7 OMs.

For the present study, the MSE and tuning of the ICES MSY rule were repeated with a shortcut MSE. The stock assessment (SAM) was replaced by an estimation emulator, which added a random error to the stock numbers at age from the OM. The stochastic short-term forecast to convert  $F$  into a catch value from SAM was replaced with a deterministic short-term forecast. Following Wiedenmann et al. (2015) and ICES (2020a), the estimation emulator error was derived from a retrospective analysis (10 years, Figure 2) and generating a log-normal distribution with a standard deviation of  $\sigma_E = 0.14$  and a lag-1 auto-correlation of  $\rho_E = 0.10$ .

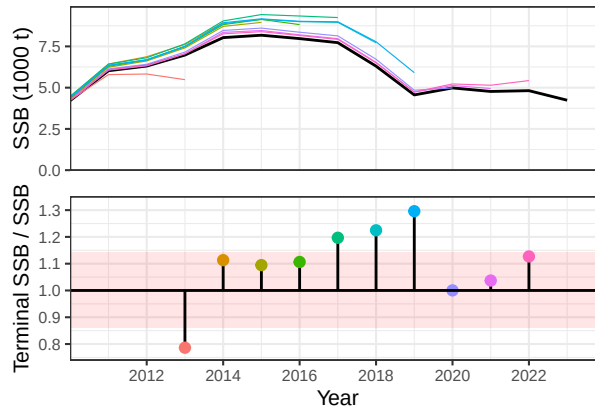


Figure 2: Illustration of stock assessment retrospective and its residuals for plaice. The red-shaded area indicates the standard deviation of the residuals.

Ultimately, the shortcut MSE kept the OM complexity of the full MSE, including the

OM reference set, but simplified the MS with a shortcut.

### 6.1.2 Results

Figure 3 shows the comparison of the tuning of the ICES MSY rule for the full MSE and shortcut MSE for plaice.

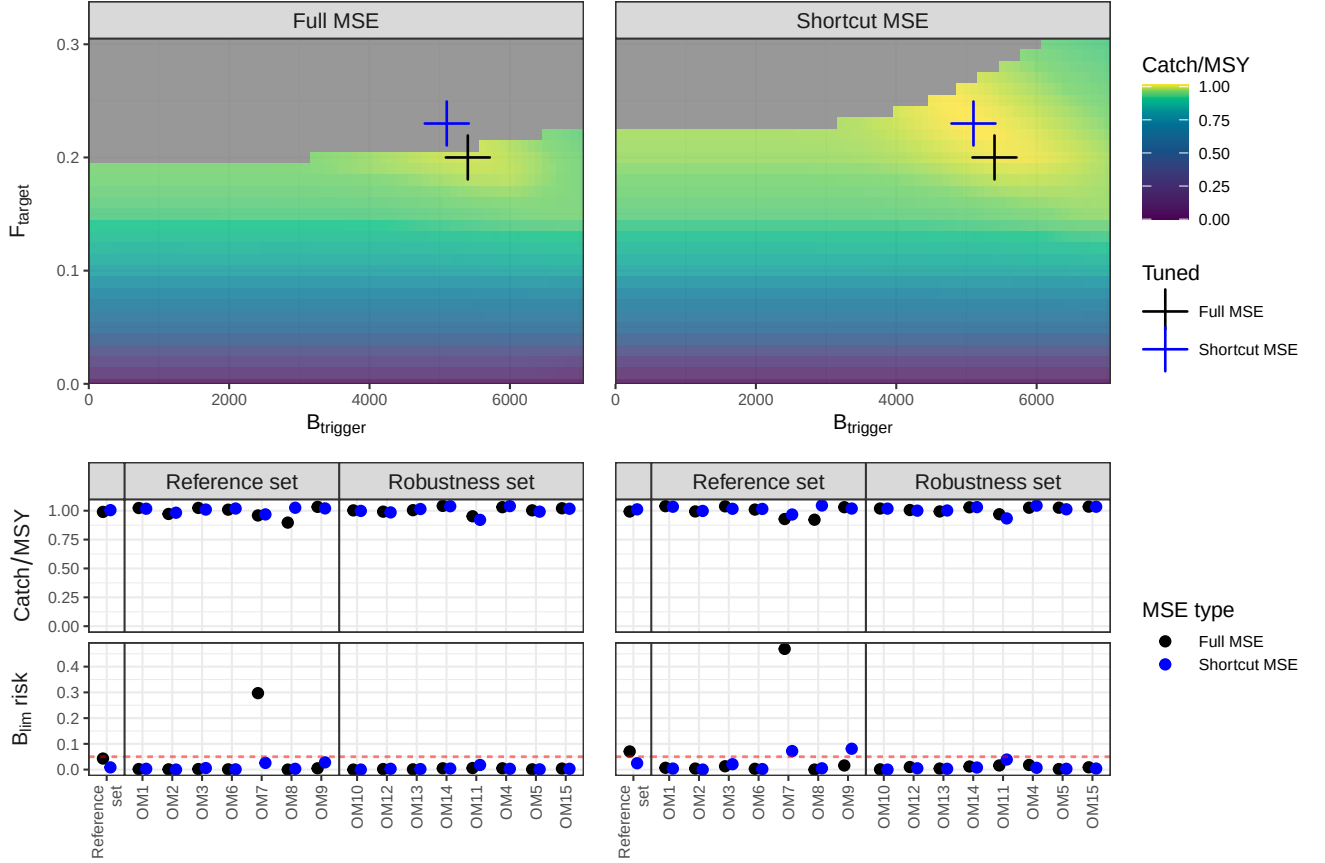


Figure 3: Comparison of results from full and shortcut MSE for plaice. The top panels show the tuning process with the OM reference set and the crosses indicate the optimum  $F_{\text{target}}$  and  $B_{\text{trigger}}$  combination. The grey shaded area indicates where  $B_{\text{lim}}$  risk exceeds 5%. The bottom panels show performance statistics by operating model (OM1–15) and for the combined reference set OM. The left panel corresponds to the tuned parameterisation from the full MSE and the left to the one from the shortcut MSE. Results from the full MSE are from Fischer (accepted).

The shortcut resulted in a optimised parameterisation where  $F_{\text{target}}$  was 15% higher and  $B_{\text{trigger}}$  6% lower compared to the full MSE and was considered non-precautionary by the full MSE because the  $B_{\text{lim}}$  risk exceeded 5%. In the full MSE,  $B_{\text{lim}}$  risk was the limiting factor. The optimised parameterisation was directly at the boundary where  $B_{\text{lim}}$  risk reaches 5% (grey area in Figure 3) and did not correspond to the total global maximum (i.e. unconstrained catch maximum). In the shortcut MSE, the risk boundary was at shifted towards higher  $F_{\text{target}}$  and lower  $B_{\text{trigger}}$  values and the optimised parameterisation corresponded to the global catch maximum. For most operating models of the OM

Table 1: Optimised parameterisations of the ICES MSY rule for plaice (Figure 3) and the three North Sea gadoid stocks (Figure 4).

| Stock   | MSE type | $F_{\text{target}}$ | $B_{\text{trigger}}$ (1000t) |
|---------|----------|---------------------|------------------------------|
| Plaice  | Full     | 0.20                | 5.4                          |
|         | Shortcut | 0.23 (+15%)         | 5.1 (-6%)                    |
| Cod     | Full     | 0.38                | 170                          |
|         | Shortcut | 0.39 (+3%)          | 160 (-6%)                    |
| Saithe  | Full     | 0.35                | 250                          |
|         | Shortcut | 0.43 (+23%)         | 240 (-4%)                    |
| Whiting | Full     | 0.14                | 220                          |
|         | Shortcut | 0.14 (+0%)          | 210 (-5%)                    |

reference set, catch and  $B_{\text{lim}}$  risk were relatively similar (Figure 3). However, larger discrepancies were observed for the operating models with alternative natural mortality assumptions (OM7–9 in Figure 3). In the shortcut MSE, the differences between the OMs were much smaller compared to the full MSE. The tuning was conducted with the 7 OMs of the reference set and in full MSE and the OM with higher natural mortality was the limiting factors because it led to a much higher  $B_{\text{lim}}$  risk (e.g. more than 10 times higher for the parameterisation optimised with the full MSE), thereby increasing the  $B_{\text{lim}}$  risk of the reference set. In the shortcut MSE, this OM resulted in a much lower  $B_{\text{lim}}$  risk, much closer to the other OMs. On the other hand, the OM with age-dependent natural mortality (instead of the age-independent baseline assumption), resulted in higher  $B_{\text{lim}}$  risks in the shortcut MSE.

## 6.2 North Sea gadoids

### 6.2.1 Methodology

In 2018–2019, ICES held a workshop to conduct an MSE for several North Sea stocks and to tune the ICES MSY rule (ICES, 2019). For all stocks, the MS was based on the SAM model. Subsequently, the MSE was repeated with a shortcut for one of these stocks, cod (*Gadus morhua*), and presented at another ICES workshop (WKGMSE3; ICES, 2020a), following the same shortcut approach as described above for plaice. For the present study, we repeated the shortcut MSE for two further stocks, saithe (*Pollachius virens*) and whiting (*Merlangius merlangus*). The estimation emulator errors ( $\sigma_E$ ) were 0.11, 0.04, and 0.17, and auto-correlation ( $\rho_E$ ) were 0.25, -0.12, and 0.47, respectively, for cod, saithe, and whiting. ICES (2019) tuned the ICES MSY rule by adjusting both  $F_{\text{target}}$  and  $B_{\text{trigger}}$  but only with a single (baseline) OM, although alternative OMs were considered for robustness tests. Due to the high computational complexity of running a full grid for a full MSE, ICES (2019) only ran a full grid of  $F_{\text{target}}$  and  $B_{\text{trigger}}$  combinations for cod. For saithe and whiting, ICES (2019) first identified the boundary between precautionary and non-precautionary parameterisations, and then focused on finding the maximum catch within the precautionary zone, resulting in sparser grids.

### 6.2.2 Results

Figure 4 and Table 1 show the results of the full MSE in comparison to the shortcut MSE.

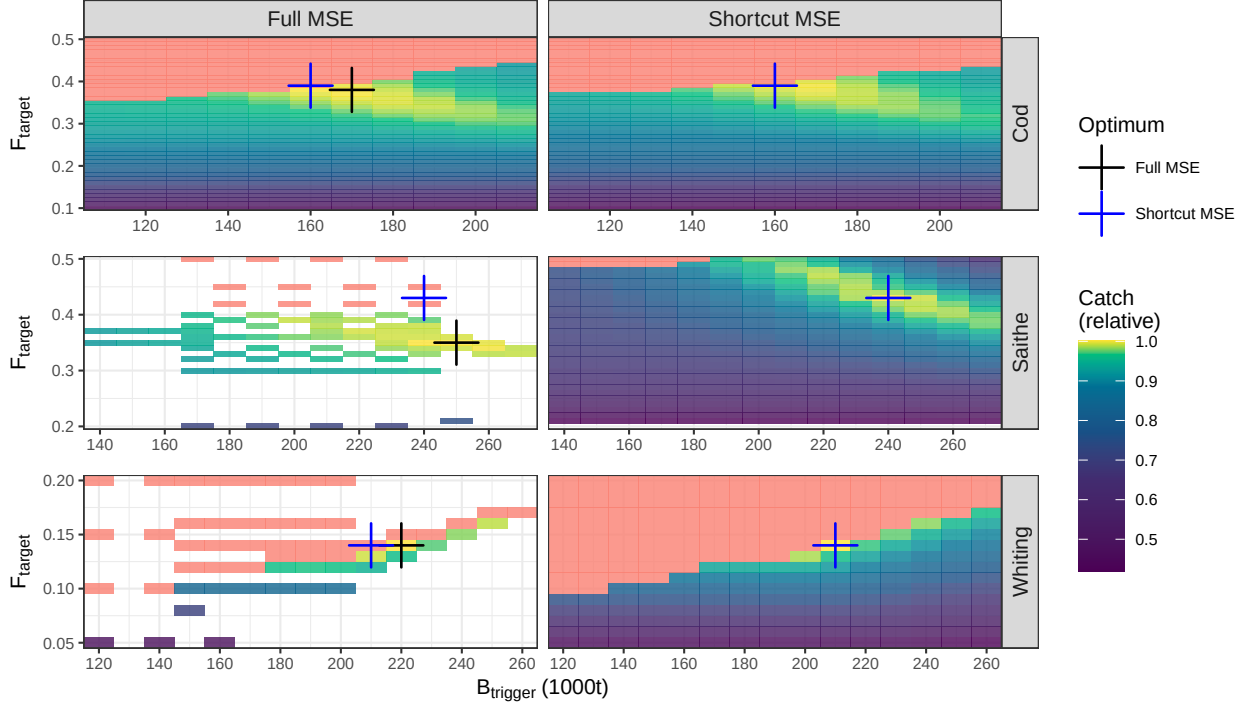


Figure 4: Results of tuning the ICES MSY rule with a full and shortcut MSE for three North Sea gadoid stocks. The red shaded area indicates non-precautionary combinations of  $F_{\text{target}}$  and  $B_{\text{trigger}}$  where  $B_{\text{lim}}$  risk exceeds 5%. The optimum is the highest precautionary catch. Results for the full MSE are taken from ICES (2019) and a full grid was only available for cod.

The shortcut MSE led to optimised parameterisation of the ICES MSY rule ( $F_{\text{target}}$  and  $B_{\text{trigger}}$ ) that were not precautionary in the full MSE for all three gadoid stocks (Figure 4). For cod and saithe,  $F_{\text{target}}$  was higher and  $B_{\text{trigger}}$  smaller, for whiting, only  $B_{\text{trigger}}$  was smaller (Table 1). The boundary where the parameters become non-precautionary was slightly shifted for cod and whiting. However, for saithe, the shortcut MSE suggested that  $F_{\text{target}}$  could be much higher (+23%) and still be precautionary (Figure 4) compared to the full MSE.

## 6.3 North East Atlantic mackerel

*NOTE: needs revising and final results*

### 6.3.1 Methodology

The North East Atlantic (NEA) mackerel *Scomber scombrus* stock is widely distributed, and management is complicated by uncertainty in stock status, differing national quotas and persistent catch overshoots (ICES, 2023a). A shortcut MSE designed to emulate

the recent assessments is compared to a full MSE. The MSE was conducted as a back test to stress test the current advice rule set up as a closed-loop retrospective simulation where the existing harvest control rule is applied. The operating model is conditioned on historical assessments and catches, and in each simulated year the MP applies the advice rule fed back into the operating model. By repeating this exercise under plausible alternative scenarios the back test evaluates the robustness of the current rule, which had been developed for a single operating model,

The MSE was based on an age-structured operating model (OM) conditioned on the recent estimates of recruitment, growth, natural mortality, and selectivity, with a reference set of OMs representing alternative assumptions about stock productivity, catch reporting, and survey dynamics. For each candidate management strategy, the management procedure included an estimation model based on the state-space assessment model SAM (Nielsen & Berg, 2014), applied to simulated monitoring data, followed by a deterministic short-term forecast to convert fishing mortality into catch advice. The ICES MSY rule (ICES, 2023a) was implemented as the harvest control rule (HCR).

For the shortcut MSE, the estimation model in the management procedure was replaced by a type 1 emulator. Instead of fitting SAM to simulated observations, the emulator took the “true” numbers-at-age from the OM and added log normal estimation error to key assessment metrics (e.g. spawning stock biomass). The error structure (standard deviation and lag-1 autocorrelation) was parameterised from a hindcast analysis of the NEA mackerel assessment over the period 2000–2020, following the approach used for deriving autocorrelated assessment error from retrospective analyses (e.g. Wiedenmann et al., 2015). In both the full and shortcut MSEs, the same OM reference set, candidate HCR grid, and performance metrics were used, allowing direct comparison of the resulting risk–yield trade-offs.

### 6.3.2 Results

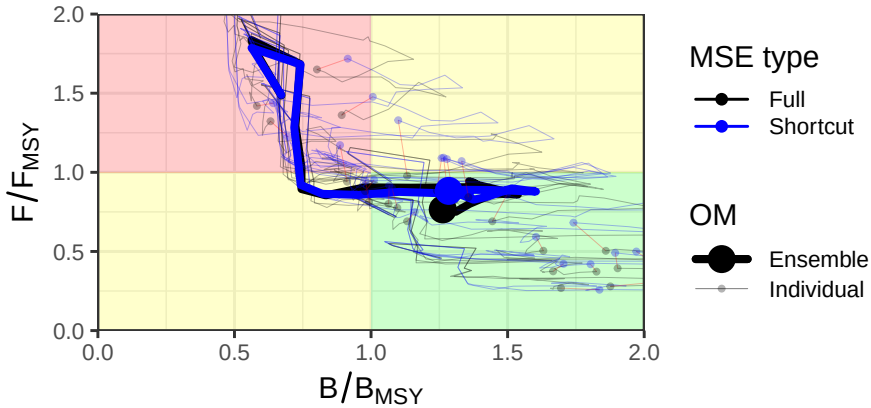


Figure 5: Kobe plot with simulation results for Northeast Atlantic mackerel. Each curve represents the 20-year time series, and the points indicate the status in the last year (2020). The results from the full and shortcut approaches are connected by red lines.

## 6.4 North Sea herring

*Benoît – add something during summer 2026 or remove from manuscript*

### 6.4.1 Methodology

### 6.4.2 Results

## 7 Discussion

### *Summary*

The main aims of this study were to define what a shortcut MSE is, compare shortcut and full MSE simulations for several case studies, and discuss the differences between the two approaches. The outcomes from the case studies showed that shortcut and full MSE results may be similar, but often are not, and relying on shortcut MSEs alone can lead to recommendations for MSs that do not meet required management objectives.

### *Tail of distribution vs risk*

One of the main objectives of evaluating MSs using MSE is often to reduce or limit the risk of stock depletion, i.e. the probability that a stock falls below a pre-determined biomass limit reference point should be low. For example, ICES generally considers management plans as precautionary if the probability of a stock falling below the biomass limit reference point  $B_{\text{lim}}$  does not exceed 5% in the long term (ICES, 2023a). Such considerations involve relatively rare events and are determined by the level and structure of uncertainty incorporated into the MSE. The Monte-Carlo nature of MSE projection means that the decision on whether an MS is suitable is often based on the tails of the simulated projections. Shortcuts can struggle to accurately emulate the level and structure of uncertainties, leading to a distribution that differs from that of a full MSE simulation and, consequently, to different conclusions regarding an MS's suitability.

### *Emulator conditioning*

It is impossible to know how well a shortcut approach emulates dynamics prior to running a full MSE. The ideal way to condition a shortcut is to first run a full MSE and then mimic its behaviour with an emulator. However, if a full MSE is run anyway, there is no need to run the shortcut in the first place. In hindcast simulations, such as the mackerel example, it is easier to condition a shortcut approach, but such approaches are more suitable for evaluating hypothetical scenarios about whether another MS may have performed better and less suitable for selecting an MS for the future. MSE is often about developing adaptive management that responds to changes in the system (e.g. regime shifts), and estimators can learn and adapt to these as the data time series increase in length. On the other hand, a shortcut MSE assumes there is inherently nothing to learn.

### *Conclusions from case studies*

The present study used several case studies to illustrate the issues arising from the use of a shortcut MSE. If a single OM is used when tuning an MS, the results can be relatively similar, as shown for cod and whiting, but also very different as was the case

for saithe. Despite the results being similar for two of the three stocks, the shortcut suggested an MS parameterisation that was not considered precautionary and therefore failed the management objectives in the full MSE for all three stocks. In situations with multiple OMs, shortcut approaches are likely to struggle to emulate the behaviour of the model estimator. This was the case in the plaice case study, where the shortcut critically underestimated the probability of being below the biomass limit reference point. Furthermore, shortcuts appear to struggle with the concept of alternative OMs. In the full MSE for plaice, the management performance of the MS was substantially different between OMs, but in the shortcut rather similar regardless of the OM. In particular, the shortcut failed to model management performance under alternative natural mortality assumptions with any accuracy. The issue here is that the shortcut MS implicitly knows about changes in the OM (e.g. a different natural mortality), whereas the model estimator does not, as would be the case in reality. This can lead to the selection of MSs in shortcut MSEs that do not meet objectives and are far riskier in practice than anticipated in a shortcut simulation.

### *Shortcut MSE types*

While not much has been published on shortcut MSEs in the scientific literature or exists as training material, some material is available on shortcut MSEs, such as a brief factsheet by harveststrategies.org (2022).

We defined three shortcut MSE types: type 1 (specific emulator), type 2 (generic emulator), and type 3 (simplified estimation model). Such shortcuts go against the recommendation that “best practice is to simulate the management strategy exactly as it would be applied in reality” (Punt et al., 2016, p. 318). The type 2 shortcut is not an MSE *sensu stricto* because the MS is not fully pre-specified (data, [estimation model], decision model). This shortcut follows rather an alternative interpretation of the MSE principle (simulation test before applying something in reality). Care should be taken when calling such simulations MSE, because the term may be misleading, particularly to non-MSE experts and stakeholders. For example, the FLR shortcut MSE currently being considered by ICES (Winker et al., 2025; ICES, 2024b) could be better described as a simulation tool for estimating (management) reference points that borrows some of the design ideas of MSE but falls short of being a comprehensive MSE. Such analyses are still useful because it is impossible to conduct stock-specific MSEs for all assessed fish stocks, e.g. ICES assesses around 270 fish stocks (ICES, 2025b). The focus of shortcut approaches is typically to reduce the complexity of the MS. OMs can also be simplified, e.g. by using a non-age-structured biomass-dynamic model as the OM, which has been used in Japan (e.g. Hamabe et al., 2024) or by the Northwest Atlantic Fisheries Organisation (NAFO, 2024).

### *Pros and cons*

The third ICES workshop on MSE guidelines (WKGMSE3; ICES, 2020a) led to a lively discussion between opponents and proponents of full and shortcut MSE and the main advantages and disadvantages for both approaches are summarised in Table 2.

Table 2: Feature comparison of full and shortcut MSE.

| Full MSE                                                                                                                                                                                                                                                                                                                                                                                                                                  | Shortcut MSE                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Advantages</b>                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Advantages</b>                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <ul style="list-style-type: none"> <li>✓ Tests exactly what is implemented (MSE best practice; Punt et al., 2016)</li> <li>✓ Easy to test MS alternative scenarios (OMs) and model misspecification</li> <li>✓ May generate plausible patterns in future assessments not seen in the past</li> </ul>                                                                                                                                      | <ul style="list-style-type: none"> <li>✓ Reduced computational complexity (faster and cheaper to run)</li> <li>✓ Easier to explore larger control parameter space and alternative MS</li> <li>✓ Easier to standardise and quality control (generic MSE tools applicable to many stocks)</li> <li>✓ Allows generic HCR testing</li> </ul>                                                                                                       |
| <b>Disadvantages</b>                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Disadvantages</b>                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <ul style="list-style-type: none"> <li>✗ High computational complexity (slow to run and expensive)</li> <li>✗ Possibly high effort to implement the estimation model</li> <li>✗ May still include some simplifications to reduce computational complexity</li> <li>✗ Often requires case-specific development</li> <li>✗ Challenge of automating a procedure for handling non-convergence of estimation model in feedback loop</li> </ul> | <ul style="list-style-type: none"> <li>✗ Does not test what is implemented (against MSE best practices; Punt et al., 2016)</li> <li>✗ Difficult to emulate the estimator behaviour (uncertainty level and bias)</li> <li>✗ Assumes estimator does not learn when more data become available</li> <li>✗ Difficult to test MS against alternative OMs (alternative scenarios and model misspecification cannot be evaluated properly)</li> </ul> |

### *HPC and smarter optimisation*

Improvements in the availability and capabilities of high-performance computing have made running MSE simulations easier and faster than ever before. In particular, the possibility to parallelise simulations, even across separate computers or compute nodes (Walker, 1992), can drastically reduce the runtime of a specific MSE scenario. Nevertheless, the computational complexity of model-based MSs can still be a limiting factor, particularly when many OMs are considered, or when tuning an MS. If an MSE needs to be tuned, it is rarely required to run exhaustive simulations with all possible combinations of MS control parameters. Instead, management objectives can be formalised as an objective function and optimisation routines can be deployed for automation, e.g genetic algorithms (Fischer et al., 2021a,b). Furthermore, statistical approaches such as history matching (Spence, 2025) can reduce the number of simulations required to find



the optimum.

### *Combine shortcut and full*

As detailed in this paper, there are clear issues with shortcut MSE approaches, and the final selection of an MS should not rely solely on the results of shortcuts. However, shortcuts can be useful in some situations and do not always need to be avoided. Instead, the advantages of shortcut MSEs can be realised when combined with full MSE in a full-shortcut-hybrid MSE. A possible three-step approach when defining and tuning MSs could be:

1. (optional) Conduct a type 2 shortcut MSE (generic emulator) to find promising functional forms for an HCR and reject those HCRs that do not work well.
2. (optional) Conduct a type 1 shortcut MSE (specific emulator) to explore a wide range of HCR control parameters to narrow down the parameter space.
3. (mandatory) Conduct a full MSE for the optimum area and make sure the final MS and HCR control parameters meet management objectives.

Ultimately, it does not matter how the final MS is found, as long as the optimum is reached and shown to be robust and to meet management objectives.

### *Alternative: empirical MS*

One may argue that conducting a shortcut MSE is preferable to no MSE. However, relying on results from a possibly flawed simulation does not help, could set unrealistic expectations, and undermine the credibility of full MSE exercises.

The ultimate way to address the question of whether or not to use shortcut MSE is to avoid the discussion and switch to empirical MSs (Rademeyer et al., 2007). Empirical MSs do not rely on an estimation model and instead use data directly in an HCR. Therefore, they are much faster to run in an MSE, and there is no need to even consider a shortcut. Empirical MSs have various benefits compared to model-based MSs, including that they are easier to test and apply and conceptually simpler and therefore easier to explain to stakeholders. Furthermore, they are applicable to a wider range of fish stocks because they can be used for both data-rich and data-limited stocks (Fischer, accepted) and can match or even exceed the performance of more complex model-based approaches.

### *Conclusion & recommendations*

We recommend exercising caution when using shortcut MSEs. They should not be used in isolation for deriving fisheries management decisions, as results may be misleading and will likely differ from the results of a full MSE. Shortcut MSEs can be a helpful supplement in combination with a full MSE, e.g. for generic simulations testing or initial explorations of MS, but the final decisions on which MS to use should always be based on a full MSE. Lastly, an alternative to arguing about whether or not to use shortcuts in MSEs is to more widely adopt empirical model-free MS.

## 8 Data Availability Statement

Input data, software code, and summarised results are available from GitHub (plaice: [https://github.com/shfischer/WKBPLAICE2024\\_ple.27.7e\\_MSE](https://github.com/shfischer/WKBPLAICE2024_ple.27.7e_MSE), plaice shortcut: [https://github.com/shfischer/WKBPLAICE2024\\_ple.27.7e\\_MSE/tree/shortcut](https://github.com/shfischer/WKBPLAICE2024_ple.27.7e_MSE/tree/shortcut), cod: [https://github.com/ices-taf/WK\\_WKNSMSE\\_cod.27.47d20](https://github.com/ices-taf/WK_WKNSMSE_cod.27.47d20), cod shortcut: [https://github.com/ices-taf/WK\\_WKNSMSE\\_cod.27.47d20/tree/shortcut](https://github.com/ices-taf/WK_WKNSMSE_cod.27.47d20/tree/shortcut))

TO DO: add links for saithe and whiting

## 9 Supplementary data

## 10 Author contributions

- Conceptualisation - José, Carryn, Simon
- Formal analysis - Simon (cod, plaice), Matt (saithe and whiting shortcut), Laurie (mackerel)
- Writing, review, and editing - Simon (lead), all (equal)

## 11 Acknowledgements

## 12 Conflict of interest

The authors have no conflicts of interest to declare.

## 13 Funding

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